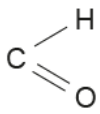
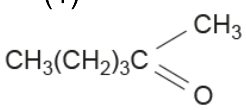
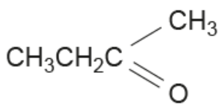
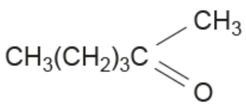
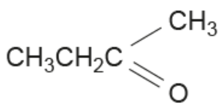
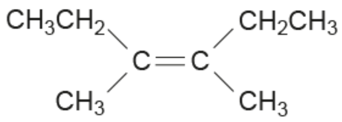
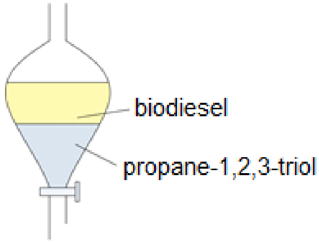
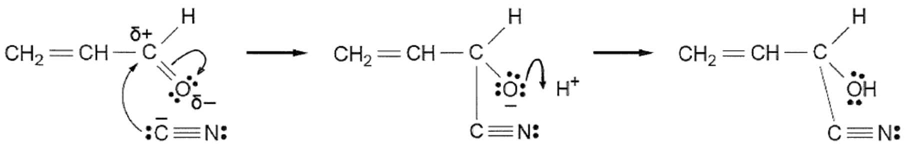
Question				Marking details	Marks available					
					AO1	**AO2**	**AO3**	**Total**	**Maths**	**Prac**
9	*(a)*	(i)		award (1) for either of following • the groups bonded to each carbon atom of the C=C must be the same • it must be a symmetrical alkene			1	1		
		(ii)		orange / red / yellow	1			1		1
		(iii)	I	award (1) for any of following • it does not contain a  group • it is not an aldehyde			1	1		
			II	it is not an aldehyde, therefore cannot be $CH_3(CH_2)_3CHO$ or $CH_3(CH_2)_2CHO$ (1) accept it must be  or  melting temperature cannot be higher than the literature value therefore it cannot be  (1) compound **U** must be  (1) ecf possible	1 1	1		3		

Question				**Marking details**	**Marks available**					
					AO1	**AO2**	**AO3**	**Total**	**Maths**	**Prac**
		(iv)		 (1) accept *E*- or *Z*- isomer 3,4-dimethylhex-3-ene (*E*- or *Z*- as structure given) (1) ecf possible from *(a)*(iii)II		1	1	2		
	(b)	(i)		award (1) for any of following • renewable source • does not use fossil fuels • method 2 gives two products (or biodiesel and propan-1,2,3-triol) • availability of raw materials	1			1		
		(ii)		award (1) each for any **two** of following • temperature conditions • pressure used • yield • rate • separation of products	2			2		2

Question				Marking details	Marks available					
					AO1	**AO2**	**AO3**	**Total**	**Maths**	**Prac**
		(iii)		award (1) for diagram **and** biodiesel labelled as top layer  award (1) for name of separating funnel	1		1	2		2
		(iv)		 partial / complete charges (1) curly arrows (1)		2		2		
				Question 9 total	**7**	**4**	**4**	**15**	**0**	**5**